COFFEE SALES DATA ANALYSIS REPORT

Executive Summary

The Coffee Sales Data Analysis project aims to transform raw sales data into meaningful business insights using Exploratory Data Analysis (EDA). In today's competitive retail industry, organizations rely on data-driven decision-making to understand customer behavior, improve operational efficiency, and maximize profitability. This report presents a comprehensive analysis of a coffee sales dataset containing approximately 10,000 transaction records.

The dataset includes valuable information such as transaction details, coffee items, quantity purchased, price per unit, total amount spent, payment methods, store locations, and transaction dates. Before analysis, the dataset contained several data quality issues including missing values, invalid entries such as **UNKNOWN** and **ERROR**, inconsistent data types, and incomplete records. These issues were addressed through a structured data cleaning process to ensure the dataset was reliable and suitable for analysis.

After preprocessing, Exploratory Data Analysis (EDA) techniques were applied to identify trends, customer purchasing behavior, product demand, sales performance, and business opportunities. Statistical summaries and visualizations helped uncover meaningful patterns that can support inventory planning, marketing strategies, customer engagement, and operational improvements.

The results of this project demonstrate how Python-based data analysis can assist businesses in making informed decisions through accurate interpretation of sales data. The project also highlights practical skills in data cleaning, visualization, statistical analysis, and business intelligence that are highly relevant for aspiring Data Analysts.

1. Introduction

Data has become one of the most valuable assets for businesses in the modern digital economy. Organizations collect large amounts of transactional information every day, but raw data alone cannot support business decisions unless it is properly analyzed. Exploratory Data Analysis (EDA) is one of the most important stages of data analytics because it helps analysts understand the structure of data, identify inconsistencies, discover hidden patterns, and generate meaningful business insights.

The Coffee Sales Data Analysis project focuses on analyzing transactional data collected from a coffee retail business. The dataset contains thousands of customer purchase records that provide detailed information about products, quantities, prices, payment methods, locations, and transaction dates. Such information can help businesses understand customer preferences, improve inventory planning, optimize pricing strategies, and increase overall profitability.

However, before meaningful analysis can be performed, it is necessary to clean and preprocess the dataset. Raw datasets often contain missing values, duplicate records, incorrect data types, and invalid entries that can reduce the accuracy of analytical results. Therefore, a complete data cleaning process was carried out before beginning the exploratory analysis.

Python was selected as the primary programming language because of its powerful data analysis ecosystem. Libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used to clean the dataset, perform statistical analysis, and create informative visualizations. These libraries provide efficient tools for handling large datasets and generating business insights.

The primary objective of this project is to understand customer purchasing behavior, identify sales trends, evaluate payment preferences, compare store performance, and provide recommendations that can help improve business performance. The project demonstrates the complete workflow of a real-world data analysis project, beginning with raw data and ending with actionable business recommendations.

2. Problem Statement

Many retail businesses generate thousands of sales transactions every day. Although this data contains valuable information, businesses often struggle to convert raw transaction records into meaningful insights. Without proper analysis, it becomes difficult to understand customer preferences, identify high-performing products, monitor sales trends, and optimize inventory.

The coffee sales dataset used in this project contained several quality issues, including missing values, invalid entries, inconsistent formatting, and incorrect data types. These problems needed to be resolved before any meaningful analysis could be performed.

This project addresses these challenges by cleaning the dataset, analyzing customer purchasing behavior, and generating actionable business insights that can support strategic decision-making.

3. Project Objectives

The primary objectives of this project are:

- To understand the structure of the coffee sales dataset.
- To identify missing values and improve data quality.

- To replace invalid values such as **UNKNOWN** and **ERROR** using suitable statistical techniques.
- To convert inappropriate data types into appropriate formats.
- To prepare a clean dataset suitable for analysis.
- To perform exploratory data analysis using Python.
- To understand customer purchasing behavior.
- To analyze quantity distribution across transactions.
- To examine pricing patterns of coffee products.
- To identify customer spending patterns.
- To compare payment methods used by customers.
- To analyze sales performance across different locations.
- To study daily sales trends.
- To evaluate relationships among numerical variables.
- To provide meaningful business recommendations based on analytical findings.

4. Scope of the Project

The scope of this project includes the complete Exploratory Data Analysis process of a coffee sales dataset. The analysis focuses on improving data quality, identifying customer purchasing trends, understanding sales performance, and generating business insights.

The project covers:

- Data inspection
- Data cleaning
- Missing value treatment
- Data type conversion
- Duplicate validation
- Exploratory data analysis
- Statistical analysis

- Business insights
- Business recommendations
- Final conclusion

The project does not include machine learning model development or predictive analytics. However, recommendations for future enhancements are provided to support advanced analytics initiatives.

5. Dataset Overview

The Coffee Sales dataset represents transactional records collected from a coffee retail business. Each row in the dataset corresponds to a single customer transaction and contains detailed information about the purchased product, quantity, pricing, payment method, store location, and transaction date.

The dataset serves as the primary source for understanding customer purchasing behavior and business performance. Since the data originates from day-to-day business operations, it reflects real-world scenarios where inconsistencies, missing values, and invalid entries are common. Before extracting meaningful insights, these issues must be addressed through a systematic data cleaning process.

The dataset provides valuable information that can help answer important business questions, such as:

- Which coffee products are purchased most frequently?
- What quantity do customers typically purchase?
- Which payment methods are most commonly used?
- Which locations generate the highest sales?
- How much does an average customer spend per transaction?
- Are there any noticeable sales trends over time?
- What factors influence total customer spending?

By answering these questions, businesses can improve inventory management, optimize pricing strategies, and enhance customer satisfaction.

Dataset Summary

Attribute	Details
Dataset Name	Coffee Sales Dataset
Total Records	Approximately 10,000 Transactions
Number of Columns	8
Data Type	Structured Transactional Data
Analysis Type	Exploratory Data Analysis (EDA)
Programming Language	Python
Libraries Used	Pandas, NumPy, Matplotlib, Seaborn

6. Dataset Description

The dataset contains multiple attributes that describe different aspects of customer transactions. Each attribute contributes valuable information for understanding purchasing behavior and sales performance.

The information available in the dataset includes product details, transaction values, customer payment preferences, store locations, and purchase dates.

A brief explanation of each column is provided below.

Column Name	Description	Data Type
Transaction ID	Unique identifier assigned to every sales transaction	Integer
Item	Name of the coffee product purchased	Object
Quantity	Number of products purchased by the customer	Integer
Price Per Unit	Price of one product	Float
Total Spent	Total transaction value	Float
Payment Method	Payment mode used by the customer	Object
Location	Store branch where purchase occurred	Object
Transaction Date	Date on which the transaction occurred	DateTime

The dataset provides sufficient information to analyze customer behavior from multiple business perspectives including sales performance, customer preferences, operational efficiency, and revenue generation.

7. Data Dictionary

A data dictionary is an important component of every data analysis project because it explains the purpose and meaning of each variable present in the dataset.

Variable	Business Meaning	Example
Transaction ID	Identifies each customer purchase uniquely	10245
Item	Coffee product purchased	Latte
Quantity	Number of units purchased	3
Price Per Unit	Cost of one unit	4.50
Total Spent	Total payment made	13.50
Payment Method	Customer payment option	Cash
Location	Store branch	Downtown
Transaction Date	Purchase Date	15-Jan-2025

The data dictionary improves dataset understanding and helps future analysts interpret variables correctly without confusion.

8. Data Quality Assessment

Before performing Exploratory Data Analysis, it is essential to evaluate the quality of the dataset.

During the initial inspection, several issues were identified.

Major Data Quality Problems

- Missing values in multiple columns
- Invalid values such as **UNKNOWN**
- Invalid values such as **ERROR**

- Incorrect data types
- Object columns representing numerical values
- Inconsistent formatting
- Possible duplicate records

These issues reduce the accuracy of statistical analysis and may produce misleading business insights if left untreated.

Therefore, a complete preprocessing workflow was implemented before beginning the analysis.

Initial Data Quality Summary

Issue	Status
Missing Values	Present
UNKNOWN Values	Present
ERROR Values	Present
Duplicate Records	Checked
Incorrect Data Types	Present
Null Values	Present
Dataset Ready for Analysis	No

9. Data Cleaning Process

Data cleaning is one of the most important phases of any data analysis project because the quality of insights depends heavily on the quality of the underlying data.

The coffee sales dataset underwent several preprocessing steps to improve consistency, completeness, and accuracy.

9.1 Importing Required Libraries

The project began by importing the necessary Python libraries.

The following libraries were used:

Library	Purpose
Pandas	Data manipulation and analysis
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Statistical visualization

These libraries provide efficient tools for handling structured datasets and generating visual representations.

9.2 Inspecting the Dataset

The dataset was initially inspected using functions such as:

- head()
- tail()
- info()
- describe()
- shape
- columns

These functions helped understand:

- Number of rows
- Number of columns
- Data types
- Missing values
- Statistical summaries

Initial inspection also revealed that several numeric variables were stored as object data types.

9.3 Handling Missing Values

Missing values can significantly affect analytical accuracy.

Different techniques were applied depending on the nature of each variable.

Numerical Columns

Missing values were replaced using:

- Mean
- Median

Categorical Columns

Missing values were replaced using:

- Mode

This approach ensured that important statistical properties of the dataset were preserved while minimizing information loss.

9.4 Handling UNKNOWN and ERROR Values

Some columns contained invalid entries such as:

- UNKNOWN
- ERROR

These values do not represent meaningful business information.

They were replaced using appropriate statistical methods including mode, mean, or median depending on the variable type.

Replacing invalid entries improved dataset consistency and prevented analytical errors during visualization and statistical analysis.

10. Exploratory Data Analysis (EDA)

10.1 Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying advanced statistical or machine learning techniques. The main purpose of EDA is to identify trends, patterns, relationships, anomalies, and potential business opportunities hidden within the data.

In this project, EDA was performed after completing all data cleaning activities. A clean dataset ensures that the generated insights are reliable and accurately represent customer purchasing behavior.

The analysis focused on understanding customer purchasing habits, spending patterns, payment preferences, product demand, and store performance. Statistical summaries were also used to understand the distribution of numerical variables and identify any unusual observations.

The major areas covered during EDA include:

- Distribution of purchased quantities
- Product pricing analysis
- Customer spending behavior
- Payment method distribution
- Store-wise sales comparison
- Sales trend analysis
- Relationship among numerical variables

The results obtained from EDA provide valuable information that can assist business managers in making data-driven decisions.

10.2 Statistical Summary

Before creating visualizations, descriptive statistics were generated to obtain an overview of the numerical variables.

The statistical summary provides information about central tendency and data variability.

Summary Statistics

Measure	Description
Count	Total number of observations
Mean	Average value
Median	Middle value
Mode	Most frequently occurring value

Minimum	Smallest value
Maximum	Largest value
Standard Deviation	Spread of data
Quartiles	Distribution of observations

These measures help identify whether the data is evenly distributed or contains extreme values.

The statistical summary also helps analysts detect unusual observations that may require further investigation.

10.3 Quantity Analysis

Quantity represents the number of coffee products purchased by a customer during a single transaction.

Understanding purchase quantity helps businesses estimate customer demand and improve inventory management.

Objectives

- Identify the most common purchase quantity.
- Understand customer buying behavior.
- Estimate inventory requirements.

Observations

- Most customers purchased between one and four products in a single transaction.
- Quantity value **3** appeared most frequently.
- Very high purchase quantities occurred less frequently.

Business Interpretation

The purchasing behavior indicates that customers generally prefer purchasing coffee products in small quantities rather than bulk orders.

This suggests that the coffee business primarily serves individual consumers instead of wholesale buyers.

Maintaining adequate stock for frequently purchased quantities can improve customer satisfaction while reducing inventory shortages.

10.4 Product Pricing Analysis

Price Per Unit represents the selling price of each coffee product.

Analyzing product pricing helps businesses understand pricing strategies and product positioning.

Objectives

- Understand product pricing distribution.
- Detect unusually high or low prices.
- Evaluate pricing consistency.

Observations

- Most products were priced within a similar range.
- Extremely expensive products represented only a small portion of the dataset.
- Product prices appeared relatively consistent.

Business Interpretation

A consistent pricing strategy improves customer trust and simplifies purchasing decisions.

Businesses should regularly monitor pricing trends to remain competitive while maintaining profitability.

10.5 Customer Spending Analysis

Total Spent represents the amount paid by customers during each transaction.

This variable directly contributes to business revenue.

Objectives

- Study customer spending behavior.
- Identify average transaction values.
- Detect high-value purchases.

Observations

- Most customers spent moderate amounts per transaction.
- High-value transactions occurred less frequently.
- Spending generally increased as purchase quantity increased.

Business Interpretation

The positive relationship between quantity purchased and total spending suggests that encouraging customers to purchase additional items could significantly increase revenue.

Businesses may introduce product bundles, discounts, or loyalty rewards to increase average transaction value.

10.6 Payment Method Analysis

Payment Method represents how customers completed their purchases.

Understanding payment preferences helps businesses improve customer convenience.

Objectives

- Identify preferred payment methods.
- Compare payment method popularity.
- Support digital payment strategies.

Observations

- Customers used multiple payment options.
- Digital payment methods contributed significantly to total transactions.
- Cash payments remained important for certain customers.

Business Interpretation

The increasing popularity of digital payment systems indicates changing customer preferences.

Businesses should continue supporting multiple payment methods to improve customer experience and reduce transaction delays.

10.7 Store Location Analysis

Location represents the branch where each transaction occurred.

Analyzing location-wise sales helps identify high-performing branches.

Objectives

- Compare store performance.
- Identify high-revenue locations.
- Support operational planning.

Observations

- Some locations consistently generated higher sales.
- Customer traffic differed across stores.
- Revenue distribution was not identical among branches.

Business Interpretation

High-performing stores may benefit from favorable locations, better customer service, or stronger demand.

Management should analyze successful branches and implement similar strategies across lower-performing locations.

10.8 Sales Trend Analysis

Sales trend analysis examines how customer purchases change over time.

Studying transaction dates helps businesses identify seasonal patterns and demand fluctuations.

Objectives

- Monitor sales over time.
- Identify peak business periods.
- Support inventory planning.

Observations

- Daily sales showed moderate fluctuations.
- Certain days experienced noticeably higher sales.
- Overall sales remained relatively stable.

Business Interpretation

Understanding sales trends allows businesses to prepare inventory before periods of high demand and optimize staffing during peak business hours.

10.9 Overall Findings from EDA

The exploratory analysis produced several important findings.

Analysis Area	Key Finding
Purchase Quantity	Customers mostly purchased small quantities.
Product Pricing	Prices remained relatively consistent.
Customer Spending	Spending increased with purchase quantity.
Payment Method	Digital payments were widely used.
Store Performance	Some branches generated significantly higher revenue.
Sales Trend	Daily sales remained stable with occasional peaks.

These findings provide valuable business intelligence that can support better operational planning and strategic decision-making.

Section Summary

The Exploratory Data Analysis successfully transformed raw transactional data into meaningful business insights. By analyzing purchasing patterns, customer spending behavior, payment preferences, and store performance, the project demonstrates how data analytics can help organizations make informed decisions.

The insights obtained from this analysis can support inventory optimization, pricing improvements, customer engagement strategies, and long-term business growth.

11. Correlation Analysis

Correlation analysis is a statistical technique used to measure the relationship between two or more numerical variables. It helps analysts understand whether changes in one variable are associated with changes in another. In business analytics, correlation plays an important role in identifying key factors that influence sales performance, customer spending, and operational efficiency.

In this project, correlation analysis was performed after completing the data cleaning process. Only numerical columns such as **Quantity**, **Price Per Unit**, and **Total Spent** were considered because correlation is meaningful only for numerical variables.

The correlation matrix provides values ranging from **-1 to +1**.

- **+1** indicates a perfect positive relationship.
- **0** indicates no relationship.
- **-1** indicates a perfect negative relationship.

Correlation Summary

Variable 1	Variable 2	Relationship	Business Meaning
Quantity	Total Spent	Strong Positive	Customers buying more products spend more money.
Price Per Unit	Total Spent	Moderate Positive	Higher-priced products increase revenue.
Quantity	Price Per Unit	Weak Positive	Purchase quantity is not strongly influenced by product price.

Interpretation

The strongest relationship was observed between **Quantity** and **Total Spent**, which is expected because customers purchasing more items naturally generate higher transaction values.

The relationship between **Price Per Unit** and **Total Spent** was positive but comparatively weaker, indicating that both price and quantity contribute to revenue generation.

No strong negative correlations were identified, suggesting that the variables generally support business growth rather than opposing each other.

12. Business Insights

The analysis generated several valuable insights that can help improve business performance and customer satisfaction.

Customer Purchasing Behaviour

The majority of customers purchased between one and four coffee products per transaction. This indicates that the business primarily serves individual consumers rather than bulk buyers.

Maintaining sufficient inventory for these commonly purchased quantities can reduce stock shortages and improve customer experience.

Revenue Generation

A clear positive relationship was observed between purchase quantity and total spending. Customers who purchased more items contributed significantly to overall revenue. Promotional campaigns encouraging customers to buy multiple products together may further increase average transaction values.

Product Pricing

Product prices remained relatively consistent throughout the dataset. This consistency indicates a stable pricing strategy that helps customers make purchasing decisions without experiencing unexpected price fluctuations.

Payment Preferences

Customers used multiple payment methods, including cash, debit cards, credit cards, and digital wallets. The increasing adoption of digital payments highlights the importance of maintaining secure and efficient electronic payment systems.

Store Performance

Location-wise analysis showed noticeable differences in sales performance across branches. High-performing stores should be studied further to identify successful operational strategies that can be replicated at lower-performing locations.

Inventory Management

Frequently purchased products should receive higher inventory priority. Proper stock management reduces lost sales opportunities and improves customer satisfaction.

13. Recommendations

Based on the findings obtained from the exploratory analysis, several recommendations can be made to improve business performance.

Inventory Optimization

- Maintain higher stock levels for frequently purchased products.
- Monitor inventory regularly to avoid stock shortages.
- Reduce overstocking of low-demand products.

Sales Improvement

- Introduce product bundle offers.
- Launch promotional campaigns during slow sales periods.
- Encourage customers to purchase complementary products.

Customer Experience

- Continue supporting multiple payment methods.
- Improve checkout speed through digital payment systems.
- Introduce customer loyalty programs to increase repeat purchases.

Business Expansion

- Study the operational practices of high-performing stores.
- Apply successful marketing strategies to lower-performing branches.
- Expand product offerings based on customer demand.

Data Analytics

- Develop an interactive dashboard using Power BI or Tableau.
- Monitor sales performance in real time.
- Automate monthly sales reporting.

Future Predictive Analytics

- Forecast future sales using machine learning.
- Predict inventory requirements based on seasonal demand.
- Build customer segmentation models for targeted marketing campaigns.

14. Limitations of the Study

Although this project successfully generated meaningful business insights, several limitations should be considered.

- The analysis was based only on the available dataset.
- Customer demographic information was not available.

- Seasonal factors and external market conditions were not included.
- Machine learning techniques were outside the scope of this project.
- Customer satisfaction and feedback data were unavailable.
- Product profitability analysis could not be performed due to the absence of cost information.

These limitations provide opportunities for future analytical improvements.

15. Future Scope

This project can be extended in several ways to provide deeper business insights.

Machine Learning

Sales forecasting models such as Linear Regression, Random Forest, or XGBoost can be implemented to predict future sales.

Customer Segmentation

Customer groups can be identified using clustering techniques such as K-Means to develop personalized marketing strategies.

Interactive Dashboards

Business dashboards can be developed using Power BI or Tableau for real-time reporting and performance monitoring.

Time Series Analysis

Historical sales data can be analyzed to identify seasonal demand patterns and forecast future trends.

Recommendation Systems

A recommendation engine can suggest coffee products based on customer purchasing history, increasing cross-selling opportunities.

16. Conclusion

This project successfully demonstrated the complete workflow of Exploratory Data Analysis using Python. The raw coffee sales dataset was transformed into a clean and structured format through comprehensive data preprocessing techniques, including handling missing values, correcting invalid entries, converting data types, and validating duplicate records.

After cleaning, the dataset was analyzed to understand customer purchasing behavior, sales performance, payment preferences, and location-wise business trends. Statistical analysis and exploratory techniques revealed valuable insights that can support strategic decision-making.

The findings indicate that customers generally purchase small quantities, spending increases with purchase quantity, and digital payment methods are becoming increasingly popular. Certain store locations consistently generated higher sales, suggesting opportunities to replicate successful operational strategies across other branches.

Overall, this project demonstrates practical skills in data cleaning, exploratory analysis, statistical interpretation, and business intelligence. These skills are essential for modern Data Analysts and can help organizations make informed, data-driven decisions.

17. References

The following resources were used as references while completing this project:

- Python Software Foundation. *Python Documentation*. https://docs.python.org/
- Pandas Documentation. https://pandas.pydata.org/docs/
- NumPy Documentation. https://numpy.org/doc/
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